

1. What is CSS? Explain the full form of CSS

CSS is a stylesheet language used to control the appearance and layout of HTML elements. HTML creates structure of a webpage, while CSS makes the webpage beautiful, responsive, and visually attractive.

2. Cascading means flowing down according to priority. When multiple CSS rules target the same element, the browser decides which rule should be applied based on the CSS cascade rules.

Ex: `h4 { color: black; }`

`h4 { color: white; }`

Output :-

white, Because the last rule overrides the first.

Real-world examples: Shopping websites

3. Cascading

- \* Cascading is the overall process CSS uses to decide which style should be applied when multiple rules conflict

- \* It considers importance, specificity and source order

- \* It decides which CSS declaration wins

Specificity

- \* Specificity is the priority/weight of a selector

It compares selectors such as ID, class and element selectors

It helps determine which selector is stronger

4. Three types of CSS

- \* Inline CSS

- \* Internal CSS

- \* External CSS

In this External CSS is recommended because CSS stored inside a separate file and it is Reusable, Easy Maintenance, professional, Faster Development

5) i) Universal Selector  
⇒ Selects every element

Syntax

```
* {  
  margin: 0;  
  padding: 0;  
}
```

ii) Element Selector  
⇒ Targets HTML tags  
Example:-

```
p {  
  color: red;  
}
```

iii) Class Selector  
⇒ Starts with .

Ex:-  
.title {  
 color: red;  
}

iv) ID Selector  
⇒ Starts with #

HTML

```
<h1 id="logo">
```

CSS

```
#logo { color: black;  
}
```

b) Class Selector

\* Used to style multiple elements

\* using a .

⇒ Ex:- .student

\* A class can be used many times on a page

7) Descendant Selector

Uses a space

Selects elements at any level inside another element

Ex:- div p

(v) Group Selector  
⇒ Multiple selectors  
Ex:- p, h1, div {  
 white, yellow;  
}

(vi) Descendant Selector  
⇒ Targets children inside another element.

```
div p {  
  color: red;  
}
```

(vii) Child Selector  
⇒ Direct child  
div > p {  
 color: blue;  
}

ID Selector

Usually used to style one unique element

using a #

Ex:- #student

An ID should be unique within a page

~~ID~~ child Selector

Uses >

Selects only direct child

Ex:- div > p

8. CSS Box Model is the fundamental concept used to understand how every HTML element occupies space on a webpage.
- Every HTML element is treated as a rectangular box by the browser. whether its: p, h1, div, img etc.
- The box model consists of 4 parts
1. Content
  2. Padding
  3. Border
  4. Margin

#### 9. Padding

Padding is the space between the content and the border.

Padding increases the internal spacing

#### Margin

Margin is the space outside the border.

Margin separates one element from another

10. box-sizing is to controls how width and height are calculated

Recommended: box-sizing: border-box;  
Width includes content, padding and border

#### 11. Block

New line  
Full width  
Width works  
Height works

#### Inline

Same line  
required width  
Width doesn't work  
Height doesn't work

#### Inline-Block

Same line  
width works  
width works  
Height works

12. Can we apply width and height to: div, span, button & why

i) `<div>` → Yes  
`<div>` is a block-level element by default we can apply width & height.

```
div { width: 200px;
      height: 100px;
      background-color: blue; }
```

ii) `<span>` → not normally.  
`<span>` is an inline element by default. The width and height generally do not affect an inline `<span>`

To make them work:

```
span { display: inline-block;
       width: 200px;
       height: 100px; }
```

⇒ Now width and height work.

- iii) `<button>` - Yes  
button is an inline-block-level element by default,  
so width and height can be applied.

```
button {  
    width: 50px;  
    height: 50px;  
}
```

- 13) Difference b/w `display: none;` & `visibility: hidden;`  
`display: none` → Hidden & No space.  
`visibility: hidden` → Hidden & Space Reserved

- 14) Flexbox in CSS, it is a layout system used to arrange elements in a row or column easily.

- i) Flex container → There are the four basic concepts of CSS Flexbox.  
The element where we write

```
• container {  
    display: flex;  
}
```

is called the flex container.

- ii) Flex items

The direct children of a flex container are called Flex items

- iii) Main Axis is the primary direction in which flex items are arranged.

`flex-direction: row;` (or) `flex-direction: column;`

- iv) Cross Axis is perpendicular to the main axis.

when: `flex-direction: row;`

\* Main axis → direction of flex items

\* Cross axis → perpendicular to the main axis

15. How do you perfectly center a div using Flexbox? Write the CSS code and explain each property.

```
• container {  
    display: flex;  
    justify-content: center;  
    align-items: center;  
    height: 100vh;  
}
```

1. `display: flex;`

makes container a flex container. Its child elements become flex items.

i) `Justify - content : center;`

Centers the flex item along the main axis with the default flex-direction: row, the main axis is horizontal.

→ Centers horizontally.

iii) `Align-items : center;`

Centers the flex item along the cross axis. with the default row direction, the cross axis is vertical.

→ Centers vertically.

iv) `height : 100vh;`

makes the container as tall as the entire browser viewport.

vh → viewport height

$100vh = 100\%$  of the screen height.